

Article

Research on the Sustainable Evolution Mechanism of Dual-Dimensional Convergence Innovation in Digital Products

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Abstract: The complex characteristics of cross-disciplinarity, dynamic expansion, ambiguity, diversity, and the rapid changes in demand significantly amplify the uncertainties in digital product innovation. The existing innovation theories, such as “stage-gate”, open innovation, agile development, and data-driven decision making, are insufficient for fully and effectively addressing these uncertainties. Based on a case study of a fintech app, we reveal that digital product innovation is similar to biological evolution, exhibiting dual life-like features of “inheritance” and “mutation” within “dual-dimensional convergence”. However, unlike natural evolution, the evolutionary process of digital product innovation can augment its use of the digital ecosystem and capabilities, establish a data-driven rapid proactive selection mechanism for the main three stages, and quickly enhance product competitiveness. The complexity of knowledge in the innovation process can be partially solved through the use of a micro-knowledge integration learning mechanism formed by the interactions of social and cognitive translation. This study also discovers that market competition and policy regulation are two unique innovation-driven characteristics in digital product innovation. This mechanism can achieve the earlier clarification of product evolution’s direction, reduce the three major uncertainties of innovation, and improve efficiency in the utilization of innovation resources to achieve sustainable development.

Keywords: digital product; dual-dimensional convergence innovation; sustainable evolution; proactive selection; cognitive translation



Citation: Weng, Z.; Cai, Y.; Weng, S.; Zuo, C. Research on the Sustainable Evolution Mechanism of Dual-Dimensional Convergence Innovation in Digital Products. *Sustainability* **2024**, *16*, 7174. <https://doi.org/10.3390/su16167174>

Academic Editors: Decui Liang, Mingwei Wang and Kobina Agbodah

Received: 1 July 2024

Revised: 17 August 2024

Accepted: 19 August 2024

Published: 21 August 2024



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1. Introduction

The digital economy is emerging as a key force in the reconfiguration of global factor resources, the reformation of the global economic structure, and the transformation of global patterns of competition. Digital products have achieved profound integration across a spectrum of domains within economic and social life, including but not limited to realms such as social networking, transportation services, financial transactions, and the tourism sector. Furthermore, the impact of digital technology on innovation performance is increasing [1], with enterprises progressively relying on digital product innovation as a means of improving their operational performance [2].

A significant number of enterprises have launched their own proprietary mobile applications, which have become key platforms for communicating with customers and creating value through the services provided. Taking the securities industry as an example, a large number of Chinese securities companies are continuously investing in mobile applications and maintaining high-speed growth [3–5]. They continue to iterate and upgrade every year. Mobile applications have transitioned from simple trading channels to comprehensive customer service platforms, with digital products’ scope continuously expanding and demonstrating ongoing integration across different sectors.

Since 1912, when Schumpeter first proposed the innovation theory, scholars have subsequently introduced concepts such as incremental and breakthrough innovation [6], disruptive and sustaining innovation [7], and autonomous, imitative, and cooperative innovation [8]. However, these classifications are relatively static and singular, failing to adequately capture the continuously evolving, life-like characteristics of digital product innovation which are akin to biological evolution. Some studies in the literature have also explored product updates and changes from perspectives such as product iteration, product lifecycle, and continuous innovation [9]. In addition, concepts such as growth products have been specifically proposed for digital products such as software, games, and digital advertising [10,11]. However, these studies have not yet encompassed the holistic characteristics of “evolution,” including the cross-fusion of different types of innovation within digital products and the continuously evolving systemic process.

In innovation process theory, one of the focal points is how to address uncertainties in innovation [12]. With the accelerated digital innovation network process, the cross-domain dynamic expansion, diversity, and ambiguity of user demands in digital products continue to enhance the complex characteristics, infinitely expanding the cognitive space required for digital product innovation and deepening cognitive models [13]. This imbues digital product innovation with more unpredictability and uncertainty [14], urgently necessitating the construction of new theoretical frameworks and paradigms that are adaptable to the complexity of digital product innovation. Traditional linear stage-gate models, which require stringent step-by-step reviews, have become significantly inadequate. Concepts such as the paradox of digital innovation and agile co-creation [15], lean startup [16], design thinking [17], open innovation [18], data-driven decision making [19,20], rapid iteration and feedback mechanisms [21], and micro-innovation [22] have emerged. These theories emphasize flexibility, user centrality, data-driven approaches, and external collaboration; however, these theories are limited by their single-dimensional perspectives and lack a systemic view. They do not originate from the foundational logic of innovation theory and therefore require a comprehensive theoretical framework that encompasses the entire process of digital product innovation. This framework should also systematically address the major uncertainties in digital product innovation—state uncertainty, effect uncertainty, and response uncertainty [23].

The digital product innovation process involves the acquisition, transformation, digestion, and integration of a significant quantity of innovative knowledge. The academic community has proposed many instrumental theories, such as establishing cross-functional teams to integrate diverse knowledge and skills for the promotion of innovation [24,25] and building knowledge management systems to collect, store, share, and apply knowledge within and outside of the organization [26,27]. However, there is limited research on the micro-level knowledge integration processes that can convert innovative knowledge into innovative decisions. Knowledge integration capability has become a critical competency for organizations to maintain competitiveness in a dynamic environment [28]. Organizations with high knowledge integration capabilities can make innovation decisions more quickly [28].

To deconstruct this complex process, existing frameworks such as the knowledge transfer theory [29], team learning and collaboration [30,31], and the situated learning theory [32] have been proposed. However, these frameworks do not systematically address the complexity of knowledge integration in digital product innovation. In innovation networks, the generation of innovative ideas stems from the continuously growing reservoir of small, diverse, and dynamic knowledge within “trading zones” [33]. Digital technologies, through their capabilities of connectivity and convergence [34], cause innovative knowledge to increasingly exhibit complex characteristics such as heterogeneity, ambiguity, volatility, and combinability. Furthermore, the semantic distance between innovative knowledge is widening, which not only enhances the diversity of innovative knowledge and resource tools but also increases the demand for the integration of this knowledge.

Therefore, understanding the different processes of knowledge integration under various innovations, especially in a digital ecosystem, and enhancing an organization's knowledge system integration capabilities to support more effective innovation decisions have become pressing issues that need to be addressed.

Innovation drivers are another focal point in innovation theory. Technological advances are regarded as the core drivers of innovation and economic development [35,36]. Market competition drives firms to continuously improve and innovate through competitive pressure and innovation incentives [35,37,38]; however, excessive competition may lead to short-term behaviors and uncertainty. Policy environments promote innovation through mechanisms such as funding support and intellectual property protection [39,40]. Customer-driven innovation emphasizes the importance of user needs and feedback in the innovation process [22,41]. In this digital era, the unique roles of digital platforms and innovation ecosystems, digital culture and industrial policies, socialized user experiences, and digital infrastructure require further in-depth analysis.

This study conducts an empirical investigation of digital product innovation in securities firms' apps within the context of China's fintech landscape. By introducing theories such as the "evolution theory" and "cognitive translation," it explores and analyzes the evolutionary process and intrinsic mechanisms of digital product innovation, examining the following three key aspects:

- (1) Compared with traditional products, what are the fundamental characteristics of the innovation and evolutionary processes of digital products, and what innovation logic should be adopted?
- (2) What is the mechanism of innovation and evolution in digital products? How does the mechanism address the uncertainties of innovation? How does it solve the complexity challenges of digital products?
- (3) What new characteristics are possessed by the influencing factors of digital product innovation?

Through case studies, a new theoretical framework for digital product innovation is constructed, enriching and deepening the theory of digital product innovation. This provides both a theoretical reference and practical insights for enterprises' digital product innovation.

2. Literature Review

2.1. Types and Basic Characteristics of Digital Products and Digital Product Innovation

Digital technologies, such as the mobile internet, artificial intelligence, cloud computing, big data, and social media, provide a significant number of tools and opportunities for the development and delivery of new digital products [42]. Digital product innovation typically manifests as new products or services achieved through either digital forms or digital technologies [13]. In a broader sense, digital products encompass digitized products such as smart wearable devices; meanwhile, in a narrower sense, they refer to pure digital products, including enterprise software and applications [43]. The digital products considered in this study specifically refer to the category of pure digital products exemplified by apps and similar entities.

Pure digital products possess inherent physical attributes, notably their non-consumable nature, ease of replication, and susceptibility to modification [44]. The convergence and generativity resulting from the physical characteristics of digital products have been recognized as the fundamental distinctions between digital products and traditional products. Convergence refers to the integration of multiple products that were formerly independent of each other, facilitated by digital technology, resulting in the provision of new products to customers. Generativity refers to the introduction of new product features, functions, and derivative innovations enabled by digital technology [45,46].

Product innovation usually involves the creation of a novel product or the enhancement of the functionality of an existing new or old product [47]. However, digital product upgrade innovation has its own unique classification dimensions and fundamental charac-

teristics. Digital products can be divided into three categories, according to their purpose and nature, as follows: content-based products, exchange tools, and digital processes and services [44]. According to the version type, they can also be divided into functional update [48], technical non-functional update [49], incremental update [50], commercial update, and hybrid update packages. According to the products' strategic performance, they can be divided into application core innovation and application support innovation [48]. Previous studies in the literature have primarily employed a binary or tripartite taxonomy following an "either/or" logic, maintaining relatively singular and static perspective and logic. However, digital products are multi-dimensional, intersecting, and inclusive entities rather than a simple aggregation of multiple dimensions. They still do not effectively capture the holistic characteristics of "evolution," such as the intersection and integration of different types of innovations within digital products and the continuously evolving systemic process. The innovation process for digital products also features an "ordered" organizational process that includes the "disordered" and chaotic, random side. There is order within the disorder, and only by combining "disorder" and "order" can the uncertainties of digital product innovation be effectively managed.

2.2. The Uncertainty and Innovation Logic of Digital Product Innovation

The greatest challenge in product innovation is uncertainty. Milliken (1987) divided innovation uncertainty into the following three categories: state uncertainty, effect uncertainty, and response uncertainty [23]. State uncertainty refers to unpredictability in understanding and forecasting the environmental state. Effect uncertainty refers to the unpredictability of the outcomes of specific actions or decisions. Finally, response uncertainty refers to the unpredictability of how an organization will respond to environmental changes. The participants in digital projects must cope with all three types of uncertainty. Uncertainty in digital product innovation can be further categorized into market uncertainty, technological uncertainty, and competitive uncertainty. The unpredictability of user needs and market trends, the rapid iteration and application of technology, and the intensity of market competition all necessitate companies to establish continuous innovation capabilities and quick response mechanisms to maintain their competitiveness [51].

Traditional product innovation has a long cycle, primarily controlling uncertainty through rigorous validation at each step. The innovation process mainly adopts a linear stage gate innovation model, where the product is promoted and optimized through a linear sequence from the initial concept to the finished product. Both the classic market demand-driven and technology-driven innovation models are linear innovation models.

However, the rapidly changing, fuzzy, and diverse demand for innovation in the digital era poses a challenge to the classic linear innovation model. The current literature has primarily proposed agile strategies and decision logic to deal with uncertainty. The significant decrease in the initial sunk costs and project conversion costs of digital product innovation has given rise to the "agile" innovation model, which emphasizes periodic iteration in the innovation process [9]. The development of products based on agile micro-services can eliminate the "digital paradox" of digital product innovation [15]. Scholars have proposed a three-stage process theory and adaptation logic for growth in digital products, which is similar to the evolution of organisms' adaptation to the environment [11]. In essence, agile logic addresses uncertainty through speed; however, the process requires substantial cost investment. Agile logic primarily addresses response uncertainty, while it is more passive towards state and effect uncertainties.

Decision logic emphasizes that companies can make more accurate decisions in product innovation through obtaining more comprehensive and detailed data information, thereby reducing innovation uncertainty. This theory emphasizes improving the efficiency of employing user information and knowledge as an important means of enhancing product innovation performance. The theory suggests exploring the impact of users on reducing innovation uncertainty from the aspects of information and knowledge acquisition, absorption, and integration [52]. The information of both leading users and ordinary users

provides an important source of data for companies' product decisions [53]. In a non-big data context, companies mainly acquire product demand and idea information through their interactions with leading users [54], and they reduce the risk of large-scale production deviations through small-scale testing [55]. In a big data context, companies primarily utilize the vast quantity of behavioral data from ordinary users to obtain comprehensive and dynamically changing demand information, thereby improving the accuracy of market trend judgments [56].

Recently, the academic community has introduced the concept of generativity, suggesting that innovation can also be achieved through the participation of a large number of heterogeneous users and information sharing among multiple entities, resulting in diversified decentralized decisions and diverse innovative outcomes to address innovation uncertainty [57]. Design thinking proposes that in addition to considering "function", one should also think about "meaning" and how to change the product framework using managerial design thinking to obtain new product insights [58]. However, in essence, this is still decision logic.

Decision logic relies on obtaining comprehensive and dynamically changing demand information. However, the cost of acquiring such information is extremely high. Moreover, the rapid changes in market demand make it difficult for companies to make accurate predictions [59]. The generative innovation model relies on the participation of a large number of heterogeneous users, but new problems such as multi-entity coordination, knowledge differences, and profit distribution make the innovation process difficult to sustain, affecting the company's innovation performance [60]. Faced with the uncertainty of product innovation in an environment of volatility, uncertainty, complexity, and ambiguity (VUCA), the academic community urgently needs to further explore new product innovation logic based on both agile logic and decision logic. In this regard, the theories of biological evolution and complex adaptive systems provide valuable insights.

Some scholars, based on Chinese practices, have proposed a three-stage growth process theory and adaptive logic for digital products. This theory suggests that, similar to biological organisms adapting to their environments, products need to make immediate adjustments based on real-time environmental feedback [11]. However, the ultra-rapid evolution of digital products is clearly not entirely akin to the slow evolution of biological organisms. Although there are similarities, significant differences from biological organisms need to be explored in order to develop more effective theories that support digital product innovation in highly uncertain environments.

2.3. Knowledge Integration and Cognitive Translation in Digital Product Innovation

In the realm of digital product development, it is crucial to effectively decipher complex and ambiguous innovative knowledge [14]. Furthermore, converting the distinct and diverse knowledge and expectations within an innovation network into new digital product attributes and requirements is a major challenge [13].

The existing literature, based on data-driven theories such as artificial intelligence (including fuzzy systems, regression analysis, and expert systems), dynamic automatic sensing, and automated mining [61,62], has achieved the replacement of human innovation decision-making experience [56].

The literature indicates that the process of integrating knowledge in digital product innovation resembles language translation, heavily depending on cognitive abilities. The cognitive translation theory aids in facilitating this integration. This perspective argues the following:

Cognitive translation in digital product innovation involves cognitive thinking activity with participants playing a central role. This includes identifying, harmonizing, assessing, and refining original innovation requirements and subsequently enhancing, transforming, and generating new innovative knowledge. During this cognitive translation process, innovation participants must translate innovation demands through interactions with a social network, a process also termed social translation [63]. Within digital innovation

networks, participants learn and exchange ideas, articulate and acknowledge each other's viewpoints, and ultimately achieve a consensus on innovative concepts. In practical settings, cognitive and social translation are intertwined, collectively advancing the amalgamation and synthesis of innovative knowledge [13]. Current research predominantly examines the determinants that influence cognitive and social translation. Nevertheless, there is a research gap concerning how these translations interact and systematically integrate throughout the digital product innovation process.

3. Research Methods

In this study, the characteristics of digital product innovation and how innovation and evolution occur are explored, and a theoretical framework for the digital product innovation process is developed. This study falls under the category of exploratory research on the innovation process, addressing the “how” question, making it suitable for a case study methodology [64]. Additionally, this research objective requires a deep data environment for the analysis of the relevant process mechanism. Conducting a single case study will facilitate the collection and in-depth analysis of detailed internal data [64].

3.1. Research Object

3.1.1. Research Situation

Amongst developing countries, China holds a leading position globally in the speed of fintech development, the level of technological innovation, and the extent of application adoption. This possesses a significant representational value and offers insights and references for research on sustainable innovation development.

The securities brokerage business is one of the industries with the highest levels of digital service capability, and it also stands out as the primary revenue source for most securities companies [3–5]. Securities brokerage services are primarily provided through mobile applications (apps), which serve as the main platform for customers. These apps are regularly updated and upgraded with each new version that is released on major app stores, accompanied by detailed app descriptions and version update logs. Consequently, a substantial amount of publicly available data are accessible for research purposes.

3.1.2. Research Object

A specific company (referred to as Company S) was selected as the case enterprise in this study. Company S is a listed securities company headquartered in China, boasting nationwide market coverage. All of its core financial indicators are at the upper-middle level in the industry, indicating strong representation in the market [3–5]. One of the authors of this paper has 15 years' experience in software development and previously served as the director of APP Development at the company. Currently, the company also has a significant collaboration with the school where the research team is based, which is beneficial for obtaining in-depth information. Since 2014, Company S has been developing a mobile securities app, boasting a 10-year history of research and development, during which the company has accumulated a substantial amount of R&D process data, and its products have undergone continuous iterations and upgrades. The app has evolved from its initial support of only the Shanghai and Shenzhen mainboard market and is now trading via a comprehensive service platform that covers all securities markets and sectors in China, integrating market data, trading, information, wealth management, and other value-added services (Figure 1). This case study complies with the principle of maintaining consistency between theoretical objectives and case selection [65], as well as the requirements of “theoretical sampling” [66].

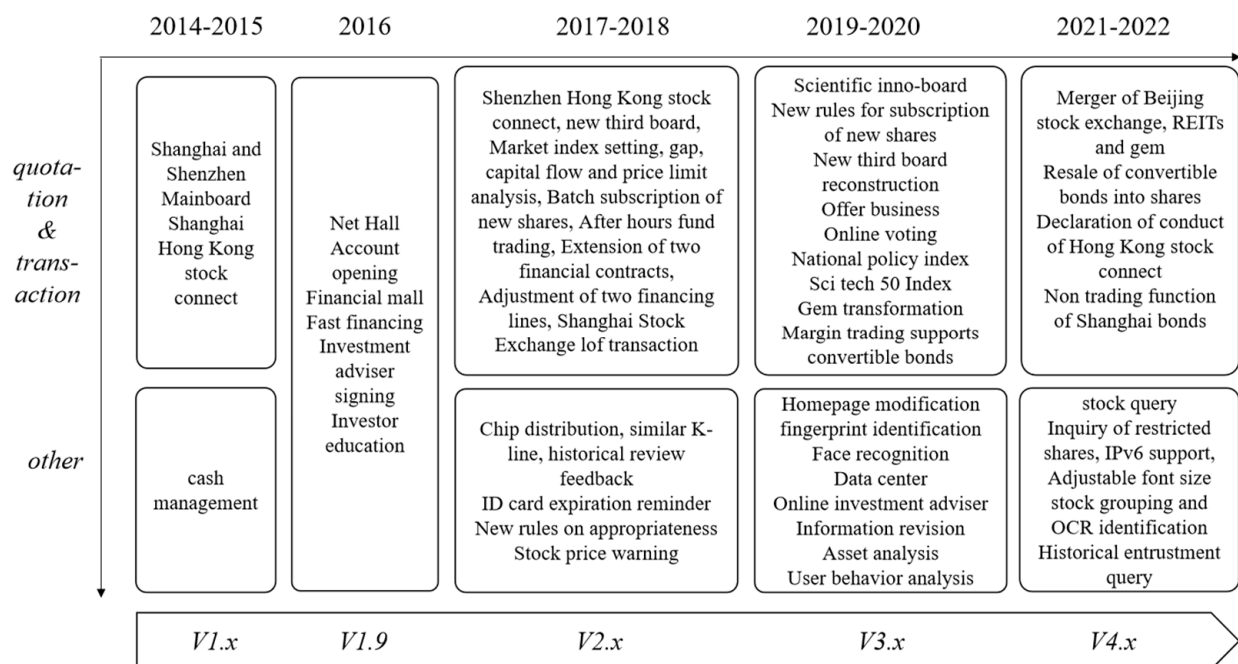


Figure 1. An overview of Company S's app development history and main product functions.

3.2. Data Collection

In this study, we used various methods to collect relevant information and data for this investigation. These methods included accessing public historical data, conducting in-depth interviews with pertinent individuals, and scrutinizing the company's internal data (such as the minutes of the product review meeting). Additionally, third-party data, such as stock exchange and Qimai data, were gathered to strengthen the evidence and achieve a triangular verification of the data [65].

The publicly available historical data from 2014 to 2023 include the contents of the company's official website, the annual reports of listed companies, official WeChat accounts, the App Store product release and customer review data, third-party platforms, etc.

Next, we obtained the necessary authorization from Company S to access the internal detailed records (40 iterations) regarding the product versions. These records were obtained through the required collection process, including demanding details from different channels, obtaining various parties' opinions during the app development process, the meeting minutes of the version review decisions, email records, requirement specifications, internal notices, promotional materials, etc.

We conducted in-depth interviews with the Chief Information Officer (CIO), the head of the technical department, two technical department experts, two project leads from the business department, and two industry third-party experts (Table 1). In this study, at least two rounds of interviews were conducted with key personnel to repeatedly confirm and supplement the experts' viewpoints. A total of 16 in-depth interviews were conducted, lasting approximately 560 min, resulting in written summaries of about 75,000 words (in Chinese). The use of multiple participants and rounds of interviews provided richer descriptions for each case and a more comprehensive understanding, helping to mitigate any biases in individual interpretations.

Table 1. The interview statistics.

Research Role	Research Department	Interviewee Position and Personnel	Number of Interviewees	Number of Interviews	Interview Duration	Audio Summary Word Count
Internal Employees of Case Company	Executive	CIO	1	2	120 min	15,000 words
	Technical	Department Head	1	2	70 min	10,000 words
		Technical Department Experts	2	4	130 min	18,000 words
	Business Department	Project Lead	2	4	120 min	16,000 words
External Experts	Senior Researcher at the Institute of Software Research, Chinese Academy of Sciences, and Chairman of a Listed Software Company		1	2	60 min	8000 words
	Product Head of China's Largest Securities Trading Software Company		1	2	60 min	8000 words

Furthermore, we gathered other data pertaining to past policy modifications within the securities industry, significant advances in smartphone operating systems and hardware, and the development framework.

4. Case Analysis

In order to thoroughly investigate the patterns of evolution in the digital products, we focused on the evolutionary characteristics of the digital products as a starting point. This stage of investigation involved dividing the stages of innovation evolution based on the dimension of time and further exploring the intrinsic logic and the key driving and supporting forces behind innovation through analyzing the mechanism of the innovation evolutionary process. The ultimate goal was to reveal a new paradigm for the theory of innovation evolution in complete digital products.

4.1. Data Analysis Method

In this study, we employed a conventional qualitative case study approach. We focused on three main aspects in relation to digital product innovation: firstly, we reviewed and summarized the basic characteristics of digital product innovation; secondly, we examined and analyzed the process that occurred; and thirdly, we identified the key factors involved in that process. The first stage primarily involved data analysis, while the second and third stages employed grounded theory to code and analyze the collected data. Additionally, visual graphic methods were used to visually represent the innovation process, and the research findings are presented through narrative descriptions, outlining the specific steps and characteristics of each stage.

The analysis process for the data in stages two and three involved several steps:

1. **Data Refinement and Open Coding:** Relevant core themes, such as innovation processes, logic, driving factors, supporting factors, requirements, channels, and methods, were screened. Two researchers independently performed open coding on the interview data to identify the initial themes. The coding results were then consolidated and discussed to resolve any inconsistencies and form a unified preliminary coding framework.
2. **Second-Level Axial Coding:** The intrinsic relationships between first-order themes were analyzed and aggregated into higher-level codes. This step aimed to unravel the complex relationships and logical structures within the data, forming second-order codes.

3. Third-Level Selective Coding: Core themes were identified and other related themes and concepts were systematically connected to these core themes. Similar second-order themes were merged and refined to ultimately form a three-level data structure [15]. 4. Theoretical Abstraction: The results from the coding of the cases were organized. Through repeated analysis, comparison, and validation, an initial theoretical framework that can explain the phenomenon was derived. 5. Data Validation: Actual case data from Company S's app innovation processes were used to repeatedly compare and verify each theme, thereby strengthening the empirical foundation of the theory. 6. Comparison of the Literature and Iterative Adjustment: The initial theoretical framework was compared with the existing literature to identify consistencies and discrepancies. Iterative adjustments and corrections were made based on feedback from the data and the literature to form the final theoretical model. The entire process employed methods such as mutual validation through interviews and data, independent work by multiple individuals, and multiple iterations to ensure the theoretical model's accuracy and applicability in practice.

4.2. Evolution Characteristics of Digital Product Innovation

We conducted a detailed analysis of the version records for the app at Company S. Additionally, the relevant information was confirmed through interviews to gain further insights into the app's underlying characteristics.

Based on the data induction and analysis, as well as the consensus among the respondents, digital product innovation could be broadly categorized into two main types, namely the optimization of existing functions and the addition of new function modules. The optimization type was designed to adapt to newly released operating systems or mobile phone features, address existing problems, and introduce new sub-functions to enhance the overall user experience.

Furthermore, we identified the addition of new functional modules as another aspect of digital product innovation. Specifically, the company expanded its business offerings through the introduction of security varieties, including the STAR market, the New Third board, Hong Kong Stock Connect, and the Beijing Stock Exchange. Additionally, the company incorporated new functions relating to financial management, investment and education, investment advisory services, account opening, stock price alerts, intelligent stock selection, etc.

4.3. Evolution Process of Digital Product Innovation

In order to comprehensively investigate the mechanism of the evolutionary process of digital product innovation, we employed the process research method to encode and analyze the interview data that were collected. The findings indicate that the innovation evolution of digital products is characterized by the iterative development of each version and can be divided into the three distinct stages (Figure 2) of demand perception, scheme formulation, and innovation implementation.

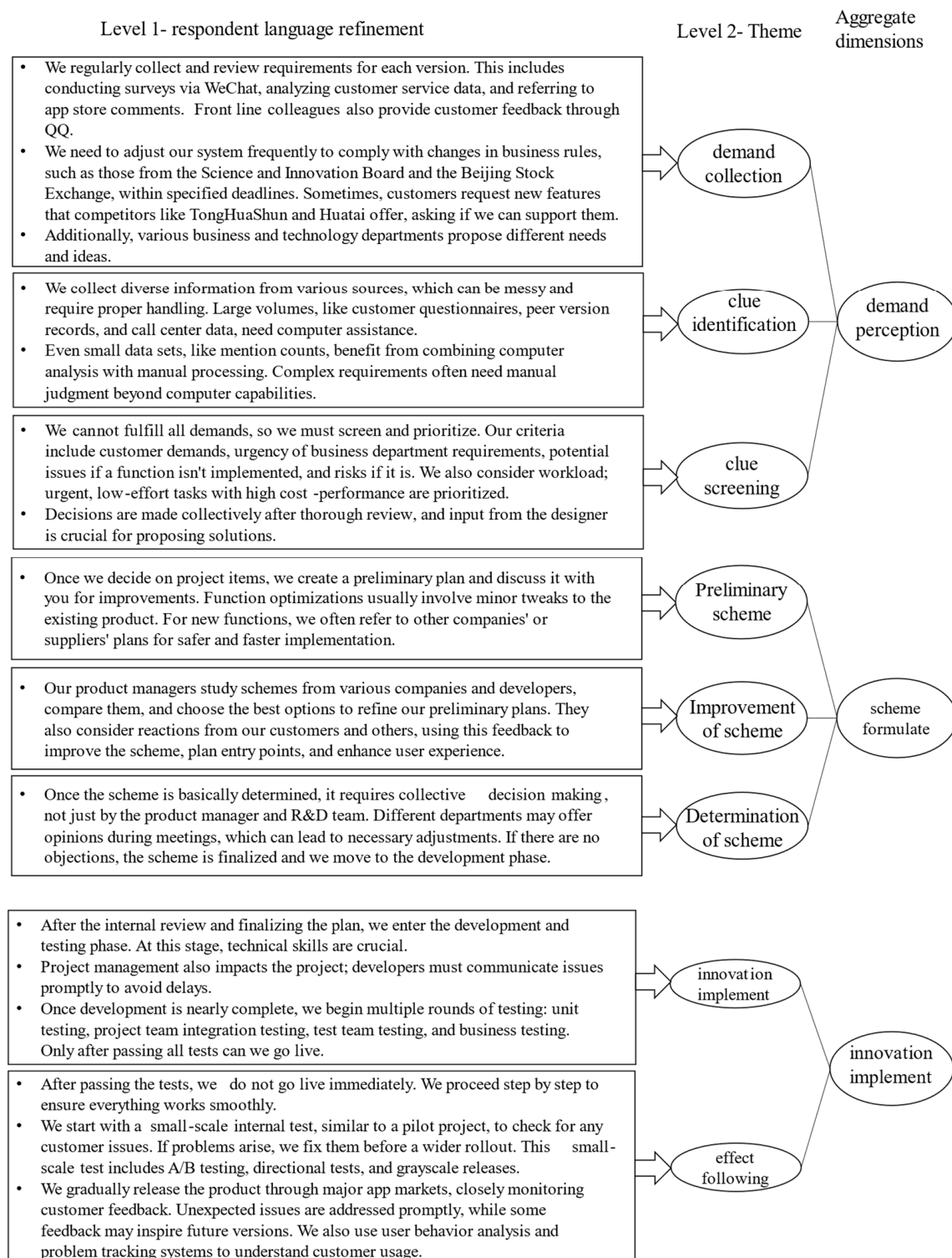


Figure 2. An example of the three-level structure of digital product innovation process research.

4.3.1. Demand Perception

In the perception stage of digital product demand for Company S, there are the following three key steps: innovation demand collection, innovation clue identification, and innovation clue screening (Figure 3).

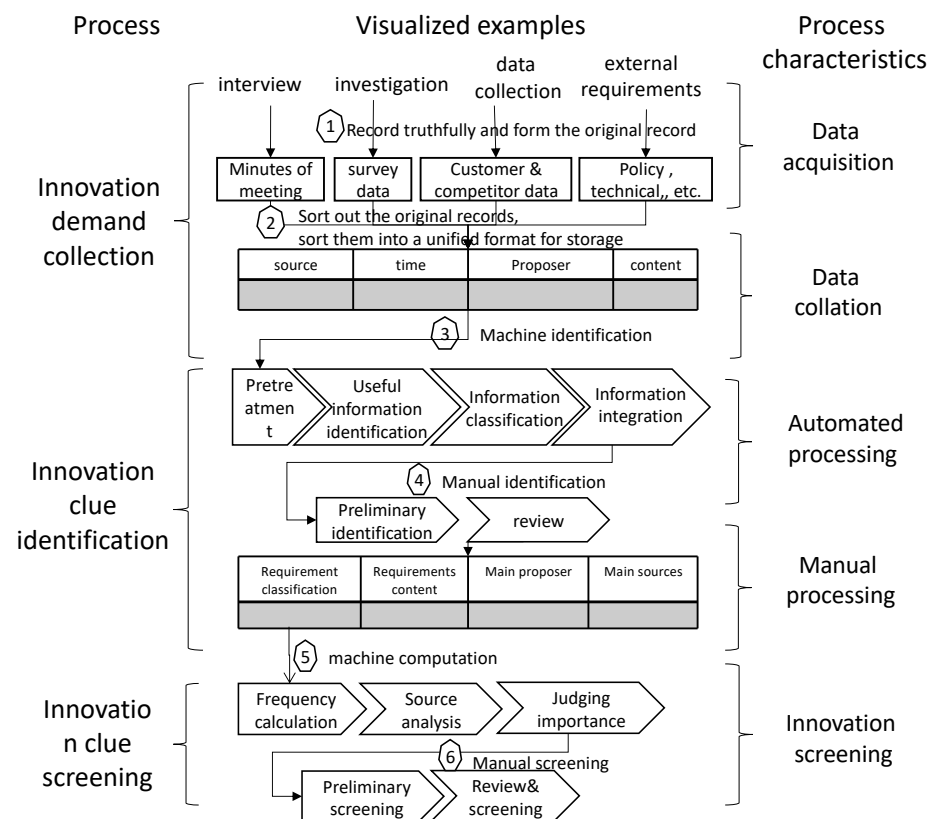


Figure 3. The perception process of digital product innovation demand.

As an example, in Version 3.0 of the company's app, the original demand data collection process is presented in Chart 1 (in logarithmic coordinates). It can be seen that the sources of demand are multi-channelled, with each channel highlighting different aspects where digital products can be improved. Missing any channel may mean overlooking some important areas that need improvement.

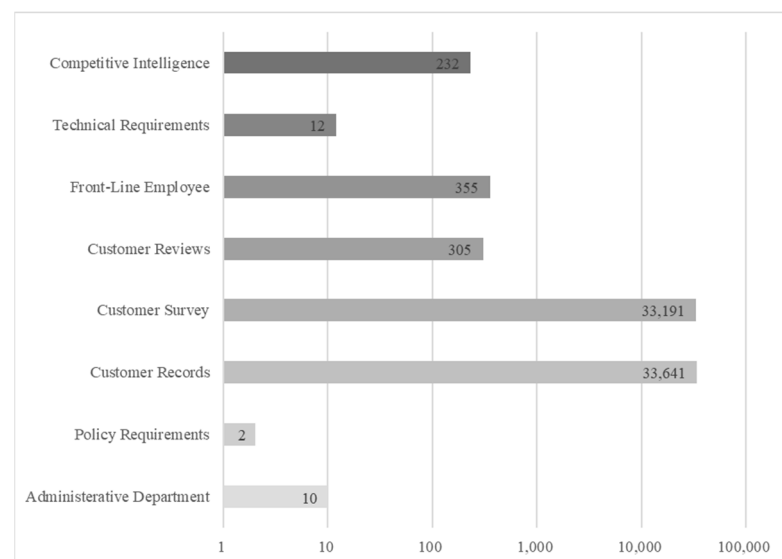


Chart 1. The original demand statistics of Version 6 (taking V3.0 as an example).

Company S primarily generates content and delivers it to customers through automated and manual processing for the purpose of value mining. The company has established a securities domain ontology to enhance the accuracy of automated processing.

In the process of innovation clue screening, numerous indicators were identified. Therefore, to facilitate the convergence of decentralized exploration, it is imperative to establish a proactive selection mechanism.

4.3.2. Scheme Formulation

The process of scheme formulation includes the three principal stages of preliminary scheme design, scheme improvement, and scheme review and determination.

(1) Improvements in the plan based on the designer's own cognition

In order to enhance the preliminary scheme, the R&D and design teams of the project, leveraging the designer's cognition, will combine their professional knowledge with the enterprise's current situation and product characteristics.

(2) Improvements in the plan based on the industry innovation database

The company has established an inter-company innovation database and has used it to improve the services provided by searching and analyzing similar innovations and customer comments.

During the innovation process of the app, all departments will collaborate to review the scheme, voice their interests and concerns, and coordinate with each other.

4.3.3. Innovation Implementation

The process of implementing innovation is divided into the following two stages:

Firstly, there is the development and testing phase, during which the R&D of innovation projects is completed and the deliverable standard is achieved. The second stage involves conducting a trial operation, the official launch, and corresponding tracking of the product.

After the product is released online, it will be periodically monitored using a user behavior monitoring and analysis system. Through the analysis of the transformation path and funnel model, it is feasible to ascertain customer preferences, identify process breakpoints, and comprehend the problems that customers might encounter while using the product. In addition, Company S has also employed performance monitoring to gain insights into customers' experiences and perceptions regarding the product's performance. In addition, real-time monitoring will be employed to promptly identify and resolve the problems encountered by online customers, including system crashes, blockages, etc.

4.4. Influencing Factors of Digital Product Innovation and Evolution

In order to further identify the key driving and supporting factors behind the evolution of digital products, we conducted an in-depth investigation of these factors in this study (Figure 4).

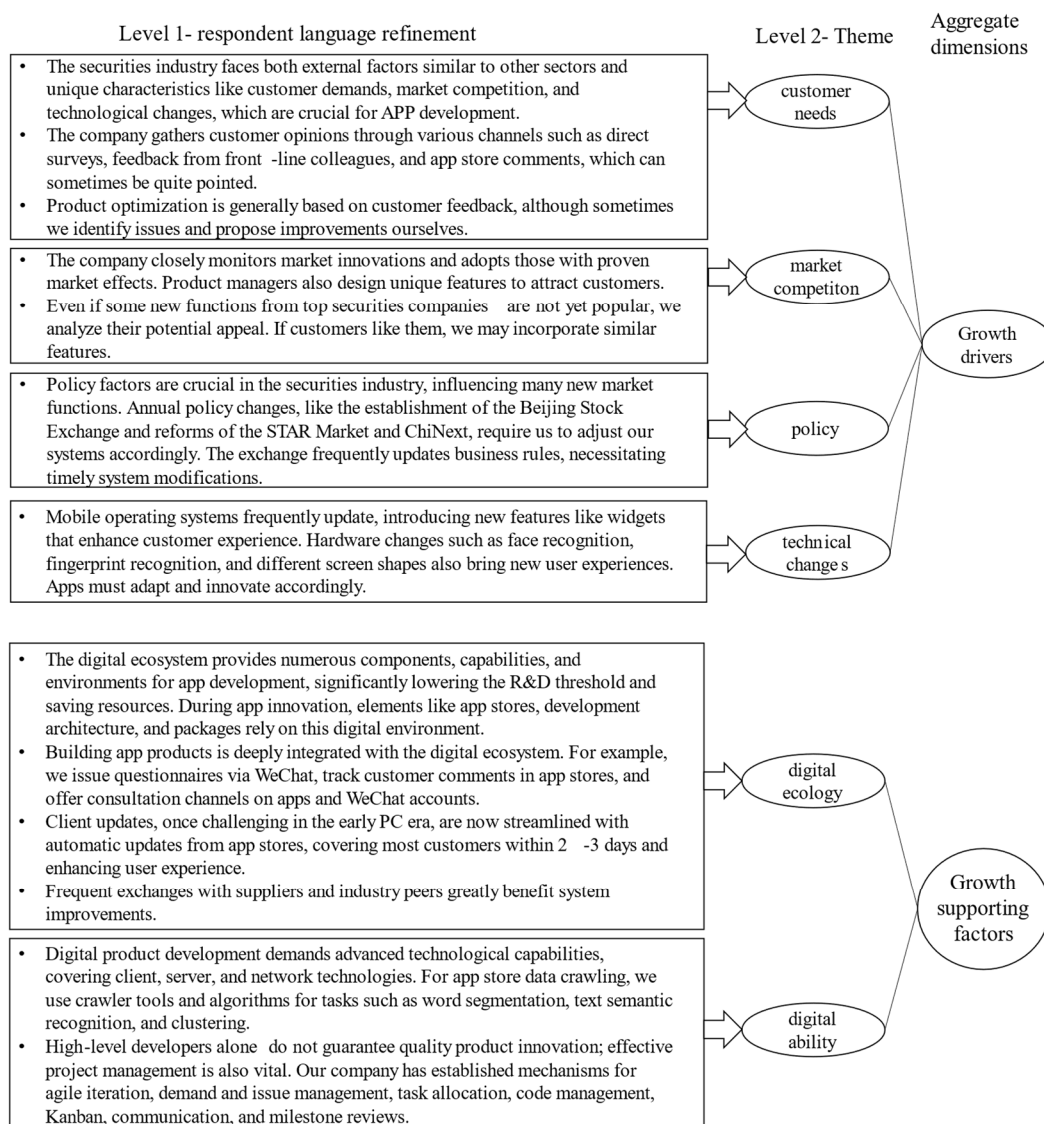


Figure 4. An example of the three-level structures of driving and supporting factors for the innovation of digital products.

4.4.1. Driving Factors

(1) Customer needs

The company will gather feedback from customers using different methods, such as direct surveys, input from employees on the front line, comments on the App Store, etc. Some customers express their opinions in a very direct and critical manner.

(2) Market competition

The company will closely monitor market innovations and proactively adopt those innovations that have been proven successful. Additionally, the product manager will take the initiative to create unique features that will appeal to customers.

(3) Technical changes

According to the survey undertaken, over the course of 8 years of digital product innovation, both iPhone and Android phones introduced numerous innovations. One notable example is the irregular screen introduced by the iPhone X, which had a significant impact on system compatibility. Additionally, in this period, both IOS and Android implemented features such as fingerprint recognition, facial recognition, and widgets.

As a result of these advances, enterprise apps have also been adapted and innovated to accommodate new features.

(4) Policy factors

The operational management department of Company S stated that the business rules of the exchange will frequently be modified; therefore, the corresponding system will need to be upgraded. This includes the STAR market and the Beijing Stock Exchange, which require upgrades to be completed within a specified time frame. The securities industry is still evolving, and numerous reform policies are typically issued each year, many of which require digital implementation.

In this study, we used the peer innovation database to confirm the driving factors of digital products. A total of 779 version records of 56 apps were collected, and Python was used to identify 10,906 innovation points. The driving factors were then divided using both computer and manual methods. Basic functions and existing business support were excluded. If a factor matched the list of historical policy reforms, it was considered to be policy-driven. The factors relating to technical characteristics were considered technology-driven.

The business strategy factors were considered market-driven. The factors matching customer comments or system improvements were considered to be customer demand-driven. The factors matching previous competitors' innovation points were considered market competition-driven. If none of these applied, the factor was classified as other. The research team analyzed the peer innovation database of Company S, and the distribution of the driving factors is shown in Chart 2.

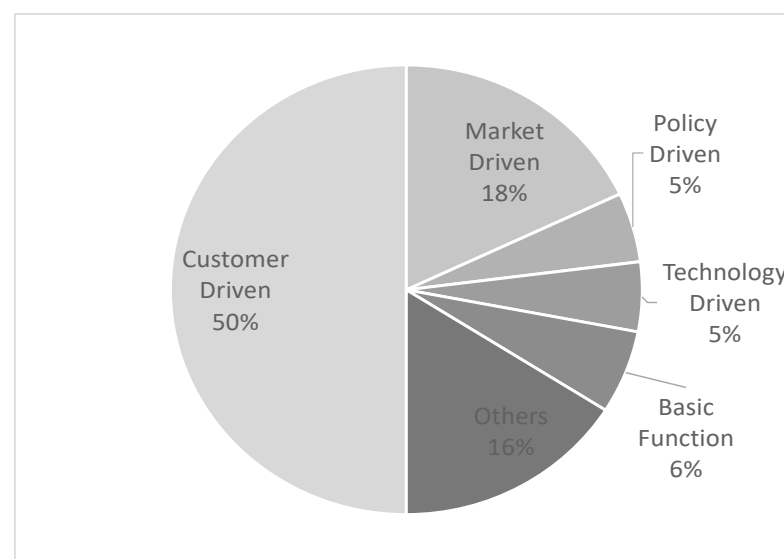


Chart 2. Analysis of driving factors.

4.4.2. Analysis of Supporting Factors

(1) Enterprise external digital ecology

Digital ecology provides a wide range of components and capabilities and an ecological environment that facilitate the development of the company's app and other digital projects. This environment significantly reduces the barriers to app development and conserves a substantial amount of the company's resources (the CIO's perspective). The construction of app products by Company S is heavily reliant on the digital ecological environment. For instance, the company frequently utilizes its official WeChat accounts to directly send questionnaires to customers. Additionally, the company's product R&D team actively monitors customer feedback on the App Store and provides channels for customer consultation and inquiries on both the app and the official WeChat account (the app's Product Manager).

(2) Enterprise's own digital capability

Firstly, the enterprise's own ability to develop digital technology is crucial. The development of digital products requires a high level of technological proficiency in areas such as client technology, server technology, and network technology (the app's Technical Manager).

Secondly, the enterprise's ability to manage digital projects is also essential. It is important to note that high-level developers do not necessarily guarantee high-quality product innovation. Therefore, effective enterprise project management is crucial. It is necessary to establish various mechanisms, such as an agile iterative development mechanism, demand and problem management mechanism, task authorization and allocation mechanism, code management mechanism, bulletin board mechanism, meeting communication mechanism, and milestone review mechanism, to ensure efficient project management (the Head of the Technical Department).

5. Theory Aggregation and Model Construction

Based on the above empirical case analysis, in this study, the logical framework method and process models are primarily employed to analyze the basic characteristics, evolutionary mechanisms, and driving and supporting factors of digital product innovation evolution. This process aims to achieve theoretical integration and the construction of a dual-dimensional convergence innovation evolution theoretical model for digital products.

5.1. Dual-Dimensional Convergence Evolutionary Feature Model for Digital Product Innovation

Scholars have proposed fundamental characteristics of digital product innovation based on different dimensions, including combinative innovation [67], iterative innovation [68], and micro-innovation [22]. These approaches primarily focused on a single-dimensional linear logic of innovation whilst still reflecting a "stage-gate" mentality and characteristics akin to tangible products. However, digital products are dual-dimensional virtual entities that can simultaneously innovate and evolve. This phenomenon is characterized by dual-dimensional convergence innovation and the evolution of digital products (Figure 5).

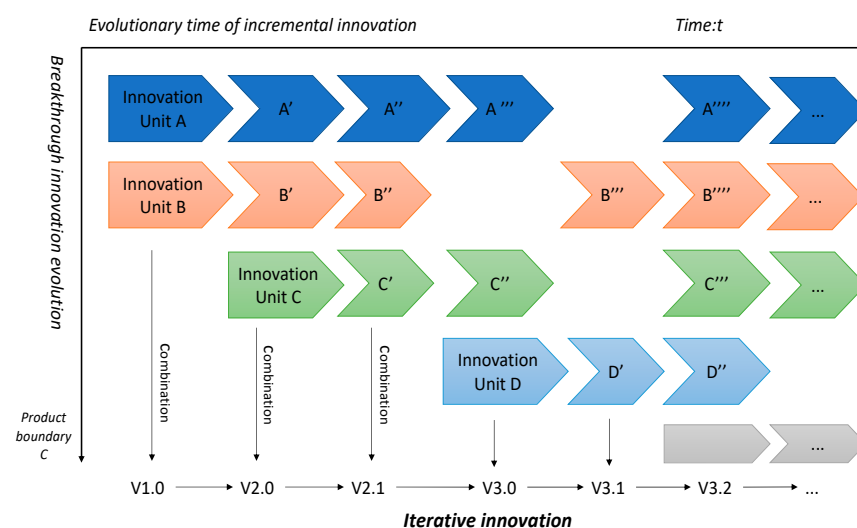


Figure 5. The dual-dimensional convergence feature model of digital product innovation and evolution.

In the context of the case study, it becomes evident that within the digital ecological environment, the process of digital product innovation has the potential not only to facilitate cost-effective upgrades and iterations, but also enable cross-border integration. In this study, we categorized digital product innovation into distinct units of innovation, denoted as U . Furthermore, the digital product, referred to as P , comprises multiple diverse innovation

units, represented as $P = \{U\}$. These units possess a certain level of independence while also relying on one another for mutual support and functionality.

The primary focus of the upgrading iteration is derived from two key aspects: first, the product boundary C , and second, the time dimension t . In terms of the time dimension t , the existing innovation units within current products will undergo iterative upgrades. This process encompasses incremental innovation, which involves problem resolution, experience enhancement, functional modification, the introduction of new sub-functions, etc. From the perspective of the C dimension of the product boundary, enterprises have the potential to achieve significant advances in their products through the incorporation of new innovation units, known as breakthrough innovation. This can be achieved by expanding their operations both upstream and downstream, as well as integrating peripheral products and venturing into cross-border markets, all while building upon their existing product offerings.

The distinctive characteristics of digital product innovation primarily lie in the interconnected integration of the time iteration and function addition. The characteristics are primarily distinguished by a dual-dimensional approach involving integrated innovation and evolution. Firstly, existing product functionalities are consistently enhanced, demonstrating a gradual and iterative evolutionary process. This is achieved through small incremental steps, utilizing low-cost trial-and-error methods. Secondly, new functionalities are also continuously incorporated, expanding the boundaries of the product, akin to the evolution and variation observed in living organisms. These two characteristics form the fundamental attributes of digital products, while their combination, iteration nature, micro-innovations, and variability are collectively externally manifested as digital product innovation and evolution.

5.2. The Digital Product Innovation Mechanism from the Perspective of the Evolution Theory

5.2.1. Three-Stage Process Model of Digital Product Innovation

Given the infinite extendibility of digital product innovation, both industry and scholars have decomposed the decision-making problems of digital product innovation into several scattered, parallel, or heterogeneous “innovation problem-solution” generation, branching, merging, termination, and refinement processes [10]. The “innovation problem-solution” proves serves as the micro-foundation for digital product innovation. The effective perception of innovation problems is a key prerequisite for innovation. Meanwhile, the design of solutions is a crucial step that determines the effectiveness of innovation. Therefore, the perception of innovation needs and problems and the formulation of solutions are all key stages in the innovation process. However, these stages primarily occur prior to the implementation of innovation, and the literature and case studies have indicated that the implementation stage of innovation is equally important. Therefore, the perception of needs, solution formulation, and innovation implementation constitute the main processes of digital product innovation.

In the needs perception phase, enterprises initially rely on digital tools to establish a collaborative network for innovation. This network is characterized by its comprehensive and rich information, diverse sources, and close interaction. An example of such a network is case enterprise v4. During the stage of perceiving demand, a total of 144 innovation clues were generated, with only 16.3% of these clues being repeated from different channels. This indicated a high level of heterogeneity in the knowledge obtained from various sources. In addition to leading customers and regular customer participation, digital innovation networks can better perceive market competition dynamics, connect partners, and build an efficient digital demand perception system.

In the solution formulation stage, within the digital innovation network, the solutions of competitors and customer feedback become more public and transparent. Companies can fully leverage this information, respecting intellectual property rights, to draw on the strengths of different competitors' solutions while avoiding elements of negative customer feedback, thereby achieving a latecomer advantage. On the other hand, leading companies

can fully utilize their market leadership role to establish core competitiveness in the market through continuous innovation.

During the implementation phase, on the one hand, there is a need to establish design at the development level. Currently, new project management or software development methodologies such as micro-services, domain-driven design, micro-kernel, adaptive and stable separation, and integrated development and operations are continually emerging based on traditional modular and robust design. On the other hand, when pushing into the market, digital product innovation can employ various mechanisms such as A/B testing, gray-scale release, and trial operation, facilitated by digital technology. These mechanisms enable the identification of system defects, data accuracy issues, and imperfect operational processes in digital products prior to their official launch.

Subsequently, through the analysis of user behavior data and the implementation of an automatic reporting system for system defects, a novel market response mode for digital products can be achieved.

5.2.2. The Proactive Selection Mechanism and Responses to Three Major Uncertainties in the Evolution of Digital Product Innovation

This study employs an ecological evolution theory approach and “adaptive logic” to deconstruct the digital product innovation process. The different innovation units are continuously engaged in random incremental exploration and thereby replace the traditional once-and-for-all “perfect design” mindset with a methodology of ongoing adaptation and optimization. The biological perspective of “evolutionary theory” primarily includes the three following aspects of genetic mutation, natural selection, and gradual inheritance. Genetic mutation provides the foundation for species evolution, while the natural selection mechanism determines whether the “traits” of mutated species are suitable for the natural ecosystem. Unlike the natural law of “survival of the fittest”, achieved through random variation and evolution in natural species, enterprises can leverage a digital ecology and their own digital capabilities to fully exercise subjective agency for “proactive selection” and planning. This allows enterprises to clarify the directions of their innovation evolution and resource allocation in advance, enhancing the efficiency of product evolution and achieving “precise” and “rapid” iterations in digital product innovation. The proactive selection mechanism plays roles in active perception, decision making, and adjustment in the three stages of digital product innovation, thereby effectively responding to the three major uncertainties of innovation [23].

The primary challenge faced by enterprises in product innovation is the uncertainty that surrounds the perception of the environmental state. Enterprises have the ability to gain real-time insights into the market state through data monitoring. Through the employment of artificial intelligence algorithms, they can identify, classify, and interpret potential innovation opportunities through the analysis of diverse, heterogeneous, and massive data. As a result, enterprises can establish a continuous and adaptable mechanism for understanding customer needs and market competition. This mechanism primarily addresses the issue of where to focus on in terms of innovation; thus, the mechanism allows enterprises to actively pursue innovation opportunities and effectively navigate the risks associated with “state uncertainty” in the innovation environment.

The second challenge encountered in product innovation is the uncertainty concerning the innovation effects. In the process of digital product innovation, enterprises can effectively integrate heterogeneous knowledge via interacting with and merging both social and cognitive translation. This involves consolidating the design teams, competitors, personnel from different departments, and feedback obtained from social media and the customer. In doing so, enterprises can anticipate customer reactions to various solutions and develop innovative approaches more efficiently and comprehensively. Consequently, enterprises can mitigate the risk of uncertainty regarding the effectiveness of their product innovation through the proactive selection of an appropriate mechanism during the formulation of strategies, thereby solving the issue of how to innovate.

The third significant challenge encountered by enterprises in product innovation pertains to the market uncertainty response. Digital products offer the opportunity for enterprises to establish monitoring technologies and mechanisms to address the market selection process, proactive customer behavior analysis, and problem collection and responses. These measures can effectively mitigate the risk of response uncertainty in digital product innovation and provide a solution for evaluating the innovation's performance.

5.2.3. A Mechanism for Integration Knowledge through the Complementary Interplay between Cognitive and Social Translation in the Evolution of Digital Product Innovation

The academic community is increasingly employing terms such as fluid or wake to depict the dynamic nature of innovative social networks, which reflects the decentralized, emergent, and other complex characteristics of digital product innovation. In essence, digital product innovation primarily generates novel ideas through the interaction and collision of social networks, knowledge, and technology. These ideas are then reconfigured, incorporating early-stage innovations and integrating crucial complementary resources and knowledge.

In the development process of solving innovation problems, the core mechanisms are founded in the integrated translation of internal knowledge resources (cognitive translation) and the integration and utilization of social resources and knowledge (social translation).

The cognitive translation process in digital product innovation is heavily reliant on the internal knowledge and experience of designers, such as either product managers or enterprises. In the case of incremental innovation, both designers and enterprises possess a deeper understanding of the existing product's functions and the associated logic of related innovation units. Meanwhile, in the process of breakthrough innovative design, this innovation may be constrained by self-cognition.

The social translation process adopts a perspective of "Interactive Meaning", which redefines the relationship between the product innovation value and the product itself. External social network resources are also employed for gathering innovative knowledge. A breakthrough innovation requires the creation of new units and the ability to surpass the boundaries of the original product definition. In the initial stages, the concept is often abstract and unclear, necessitating the input of a diverse range of knowledge to transform the abstract concept to a specific solution via social translation. Subsequently, the designer team will adapt and optimize the solution, considering the enterprise's resource status, business model, business process, customer preferences, and other cognitive translation mechanisms.

Through the collaborative interplay of cognitive and social translation that we can enhance our understanding of the dynamic nature of innovative knowledge and transform it into clear innovation cues. Subsequently, these cues enable us to develop innovation strategies that effectively incorporate both experiential insights and lessons learned. In this context, the convergence of cognitive and social translation serves as the micro-level mechanism for integrating knowledge into the realm of digital product innovation.

5.3. External Factors Affecting the Innovation and Evolution of Digital Products

5.3.1. Driving Factor

The consensus among previous studies is that product innovation, market demand, and technology are interconnected factors. Since the introduction of the service-led logic by Vargo and Lusch [69], it has been recognized that customers not only act as the source of market demand but can also contribute a wealth of diverse innovative knowledge and resources to enterprises. Furthermore, the advent of digital technology has made it feasible for ordinary consumers to participate in R&D and innovation at a low cost.

The advent of the digital economy has expedited the accessibility and clarity of market information, thereby exerting significant pressure on innovation through industry and cross-border competition. This phenomenon is particularly evident in the securities industry, where a securities firm that achieves a breakthrough innovation and garners a

positive market response will swiftly be emulated by other firms, expediting the iterative enhancement of digital products.

Government regulations and policy incentives and constraints play a crucial role in shaping the landscape of enterprise digital product innovation. These policies serve as drivers and guides for the development and advancement of digital products. For example, the introduction of new securities businesses and regulations via the exchange has significantly expanded the scope of digital products for enterprises. Conversely, the government has also implemented measures such as privacy protection and management optimization to address emerging digital challenges, such as data security and responsibility alienation, in the digital financial sector.

5.3.2. Supporting Factor

In the process of innovation, traditional methods rely primarily on an enterprise's internal R&D capabilities, as well as the support from its upstream and downstream industrial chain. On the other hand, digital product innovation is predominantly driven by the digital ecosystem and the enterprise's digital capabilities.

The characteristics of digital innovation allow enterprises to utilize digital connectivity tools to build their own digital innovation networks. These networks allow for the integration of innovation resources and facilitate the examination of digital product innovation from the perspectives of multiple participants. Through interaction and collaboration among the participants, a dynamic innovation environment and a system are created. This approach enables the free flow of heterogeneous knowledge, which expands the scope of digital innovation and enhances enterprises' ability to understand societal, customer, and industry trends. Additionally, digital ecology provides various public products, such as digital tools and components, market test environment, and release platforms, which further encourage organizations to pursue innovations.

Unlike the resource-based view and dynamic capability theory, which have clear boundaries within the traditional product economy paradigm, the ability of enterprises to establish digitalization primarily relies on the interaction and convergence between technology and business departments, customers, branches, suppliers, and other stakeholders. Enterprises must transcend the organizational boundaries and establish an internal and external resource integration system so that they can adapt to the digital era. This system enables the sharing of resources and knowledge within the innovation network. The digital capabilities encompass not only the digital infrastructure and big data analysis capabilities but also emphasize the achievement of market awareness, real-time response capabilities, and digital governance capabilities through digital connectivity and convergence.

5.4. *An Overall Model for the Evolution of Digital Products*

5.4.1. Overall Model

In this study, based on the above analysis and research, we present an integrated theoretical model of digital product innovation evolution. This model encompasses various aspects, including the driving factors of digital product innovation, the mechanism of the innovation process, the integration of innovative knowledge, and the supporting factors for innovation. This model is visually represented in Figure 6.

In contrast to the traditional dichotomy observed in product innovation, the innovation of digital products is characterized by integration and inter-transformation, resembling the Chinese Tai Chi pattern. In this study, we present a dual-dimensional convergence innovation evolution model. This model encompasses the stages of demand perception, scheme formulation, and innovation implementation. The innovation process involves the identification of clues relating to existing product improvement and breakthrough innovation, leading to the development of distinct innovation strategies that ultimately converge into a unified digital product version. The evolutionary processes of incremental and breakthrough innovation intertwine to facilitate the overall evolution of digital products. Simultaneously, the synergistic combination of cognitive and social translation

enables the realization of an extensive, diverse, and ambiguous perception of demand and the formulation of strategies. Within the three-stage process of innovation, the proactive selection mechanism effectively addresses the uncertainties that are associated with innovation, encompassing mechanisms for perceiving and identifying active demand, schemes integrating those mechanisms, and an intelligent monitoring mechanism.

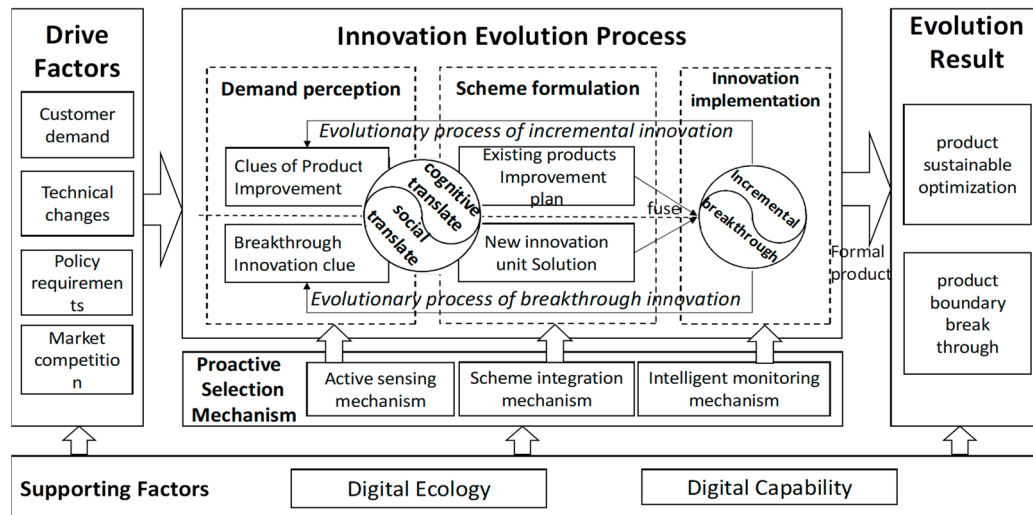


Figure 6. The dual-dimensional convergence innovation evolution integration model of digital products.

The driving forces behind digital product innovation primarily consist of external factors such as customer demand, market competition, policy requirements, and technological advances. Furthermore, the support of a digital ecosystem and the development of a company's own digital capabilities are also essential.

5.4.2. Comparison of Traditional Product Innovation and Digital Product Innovation

The innovation and advancement of digital products is different from the innovation of conventional physical products because it encompasses not only the ongoing enhancement and expansion of product boundaries, but also the distinctive characteristics, processes, mechanisms, and methodologies of innovation (Table 2).

However, there are differences in the types of innovation. Traditional product innovation mainly manifests as either incremental innovation or breakthrough innovation. In contrast, digital product innovation demonstrates an intersection of these two types, encompassing both incremental iterative optimization and breakthrough functional expansion and cross-boundary integration [70]. This dual-dimensional convergence method allows digital products to simultaneously undergo evolution and optimization across different dimensions, thereby enhancing their adaptability and competitiveness.

In terms of the innovation model, traditional product innovation relies on a linear innovation model that typically follows fixed R&D processes and stages [71]. On the other hand, in digital product innovation, a dynamic innovation model is often adopted that emphasizes multi-dimensional and multi-level innovation interaction and integration. Through the digital ecosystem, digital product innovation integrates internal and external resources, manifesting as an open innovation network [72]. This difference in resource relationships enables the producers of digital products to better perceive society, customer needs, and industry trends.

As for innovation drivers, traditional product innovation mainly emphasizes the dual factors of market demand and technological advancement [35]. In comparison, digital product innovation is more influenced by customer feedback, social networks, the digital ecosystem, and the industrial policy environment [69]. The digital economy accelerates

the flow and transparency of market information, and the pressure from cross-boundary competition acts as a significant new driver for innovation.

Table 2. A comparison of the traditional innovation paradigm and dual-dimensional convergence innovation evolution paradigm of digital products.

Type of Innovation	Traditional Product Innovation		Digital Product Innovation
Innovation paradigm	Dichotomy Linear Stage Gate Innovation		Dual-dimensional convergence evolutionary
	Incremental innovation	Breakthrough innovation	
Existing product relationships	Based on existing products	Little physical correlation	Innovation based on existing product entrance
Degree of innovation	Existing unit upgrade iteration	New innovation unit	Upgrade iteration/add
Innovation frequency	Frequently	Longer time	Frequently
Process characteristics	Linear discontinuous process	Nonlinear discontinuous process	Nonlinear continuous process
Innovation effect	Improvement and optimization of existing products	Expansion of existing product boundaries	Product improvement/boundary expansion
Demand interpretation method	Focus on cognitive translation	Focus on social translation	Interactive integration of cognitive and social translation
Scheme characteristics	Microenterprise upgrade of existing products	New innovation unit	New innovation/microenterprise iteration

The complexity and dynamism of the innovation process are the other prominent features of digital product innovation. The process of traditional product innovation is relatively linear, generally including stages such as need identification, R&D design, production, and market promotion [73]. Conversely, the digital product innovation process is more complex and dynamic, covering stages such as need perception, solution formulation, and innovation implementation, which can be quickly iterated and optimized using digital tools.

Regarding the innovation mechanism, traditional product innovation mainly relies on internal R&D mechanisms and management processes [38]. However, digital product innovation integrates diverse knowledge and resources through the interaction of cognitive and social translation to form an innovation mechanism. Cognitive translation principally depends on the knowledge and experience of designers and internal enterprise resources; meanwhile, social translation emphasizes drawing on heterogeneous knowledge from external sources [74].

Lastly, the support for innovation differs. Traditional product innovation is primarily supported by the enterprise's own R&D strength and the upstream and downstream industry chains [75]. In contrast, the support for digital product innovation mainly comes from the digital ecosystem and the enterprise's digital capabilities, including digital infrastructure, big data analysis capabilities, and real-time responsiveness [76].

In conclusion, compared to traditional product innovation, digital product innovation places a greater emphasis on multi-dimensional integration, open resource integration, and dynamic innovation processes, adapting to the rapid changes and complexity requirements of the digital economy era.

6. Conclusions and Implications

6.1. Discussion

6.1.1. The Linear Perfection Logic of Traditional Innovation vs. The Dynamic Evolution and Adaptability Logic of Digital Product Innovation

Compared to traditional physical product innovation, digital product innovation exhibits new characteristics. Traditional product innovation and R&D primarily target relatively closed environments where the problems faced are also relatively closed, allowing for relatively perfect innovation methods. On the other hand, digital product innovation

confronts an entirely new, open, and continuously fast-changing dynamic problem model, challenging the logic of perfect innovation.

Darwin proposed the theory of evolution 160 years ago, through empirical research, discovering that species evolve gradually over time rather than being precisely designed from the start. The development of species is influenced by external environmental competition. To preserve information, species replicate vast amounts of genes to respond to changing environments, creating redundancy within the population. Genes also undergo mutations, which have random and divergent characteristics, leading to differences among species. Then, natural selection allows certain “important and environment-adaptive” traits to be preserved and survive, thus realizing the “survival of the fittest”.

This is very similar to digital product innovation. Digital products also engage in exploratory innovations in various directions, akin to “gene mutations”, through different innovation units (such as Minimum Viable Products (MVPs) and A/B testing) to test market reactions. If these innovations achieve the desired market effect, the corresponding features will be “inherited” and incrementally optimized in subsequent innovations. Otherwise, they will be eliminated or subjected to further “mutations” before undergoing another round of testing. Therefore, in a certain sense, digital products exhibit bio-like characteristics, and adopting an adaptive logic in the innovation process is reasonable.

Of course, this does not mean that the quest for perfect innovation is incorrect. On a micro or technical level, digital product innovation still requires thoughtful consideration and handling based on the philosophy of perfect design. However, it necessitates an open and inclusive mindset.

6.1.2. Similarities and Differences between Digital Product Innovation Evolution Theory and Existing Related Theories

In this study, the viewpoint of the digital product innovation evolution theory also exhibits significant differences from the classic “theory of evolution”, specifically that digital product innovation experiences rapid and dynamic changes, whereas the evolution of a natural species is relatively prolonged. The essential difference between active choice and passive adaptation lies in the fact that passive adaptation involves passively accepting natural selection, while active choice entails enterprises proactively perceiving the market environment, identifying potential risks and needs, leveraging the ingenuity of both the organization and its personnel, and seizing the initiative in market competition.

To quickly establish a market position, digital product innovation involves substantial cross-domain knowledge acquisition. Although animals in nature also undergo learning or adaptation processes, these are on a lower level. In fact, learning is a key survival skill for humans on Earth, and organizations, as entities, operate similarly. Therefore, in this study, it is posited that the mechanisms of active choice and the integration of micro-level knowledge learning are the keys that distinguish the digital product innovation evolution theory from the natural evolution theory.

The innovation evolution mechanism in this study actually integrates various innovation concepts, including open innovation, rapid iteration and feedback, data-driven decision making, design thinking, and micro-innovation, demonstrating a holistic approach.

Open innovation emphasizes sourcing ideas and technologies from external sources through combining external resources with internal innovation. Both share high sensitivity and adaptability to the external environment. While the digital product innovation evolution theory also values external feedback and market responses, open innovation places greater emphasis on inter-organizational collaboration and the sharing of open resources.

Rapid iteration and feedback mechanisms emphasize the continuous improvement of products through short cycles of development, testing, and user feedback. Like the digital product innovation evolution theory, they focus on quickly responding to market changes and user needs. However, the former is more of a methodology, while the digital product innovation evolution theory likens iteration to biological evolution, emphasizing the selection and optimization of adaptive traits.

Data-driven decision making guides business decisions through analyzing data, emphasizing scientifically based decisions. Their connection with the digital product innovation evolution theory lies in their reliance on data and feedback to guide improvements and optimizations. The difference is that the digital product innovation evolution theory further emphasizes achieving product evolution through exploratory innovations and environmental selection rather than solely through the scientific nature of decision making.

Design thinking is a human-centered innovation approach that improves on innovation by discovering needs, defining problems, generating ideas, designing prototypes, and testing. Its link with the digital product innovation evolution theory is the emphasis on user feedback and iterative improvement. The difference is that design thinking focuses more on the creative process and user experience, while the digital product innovation evolution theory focuses on the process of adaptation and evolution in dynamic and uncertain environments.

Micro-innovation stresses the continuous improvement of products and services through small-scale, incremental innovations. This aligns well with the incremental improvement and adaptive logic of the digital product innovation evolution theory. The difference lies in micro-innovation's focus on minor, localized improvements, while the digital product innovation evolution theory not only focuses on incremental improvements but also emphasizes diverse trials and the selection of traits that adapt to the environment.

Overall, the digital product innovation evolution theory shares similar characteristics with these existing innovation theories, including an emphasis on rapid response, data-driven approaches, user feedback, and incremental improvement. However, it also offers a unique perspective, particularly by likening the innovation process to biological evolution. This approach highlights adaptability, variation, and selection mechanisms. This framework better reflects the dynamic competition and survival strategies of digital product innovation in an open environment.

6.2. Research Conclusion and Theoretical Value

In this study, through a multi-dimensional analysis, the characteristics, processes, mechanisms of digital product innovation, and its distinctions and connections with traditional product innovation were systematically explored. The main conclusions are detailed below.

6.2.1. Characteristics of Digital Product Innovation Evolution and Adaptive Logic

In this study, it was discovered that digital product innovation exhibits a dual-dimensional convergence evolution characteristic distinct from traditional products. Digital product innovation is not only manifested as incremental iterations over time—known as “genetic evolution” (such as problem fixes and experience optimization)—but also as breakthrough innovation in the product boundary dimension—known as “gene mutation” (such as new features and cross-domain integration). These two aspects intersect and converge, thereby continuously expanding the boundaries of the product and optimizing existing products. Additionally, digital product innovation is characterized by rapid iteration over time and the nested combination of innovation units. Digital product innovation can evolve simultaneously from different dimensions and should not be confined to the static research and analysis of a single dimension.

Digital product innovation is similar to the evolutionary process of a “living organism” and therefore should adopt the evolutionary adaptation logic as its guiding innovation philosophy. The adaptive logic of digital product innovation is reflected in its dual-dimensional convergence innovation characteristics. Specifically, digital products achieve a combination of incremental optimization and breakthrough innovation through an iterative synchronization across both the time and product boundary dimensions [70]. This integrated innovation approach allows digital products to flexibly respond to rapidly changing market demands and technological advances, thus enhancing the product's adaptability and competitiveness.

The dual-dimensional convergence evolution characteristics model proposed in this study identifies the key features of digital product innovation—the characteristics akin to the evolution of living organisms. Adaptive logic should be considered as the essential guiding philosophy for digital product innovation.

6.2.2. The Process of Digital Product Innovation and Coping with Innovation Uncertainty

This paper divides the digital product innovation process into the three stages of demand perception, solution formulation, and innovation implementation.

During the demand perception stage, market dynamics and customer needs are effectively perceived through a digital collaborative network. In the solution formulation stage, an open digital innovation network is leveraged to absorb the feedback from competitors and customers, thereby devising better innovation solutions. In the innovation implementation stage, methods such as A/B testing and phased rollout are adopted to achieve rapid iterations and precise optimizations of the digital products.

In the three-stage theoretical model, it is suggested in this paper that the characteristics of digital product innovation involve proactive selection and the integration of innovative knowledge based on cognitive translation. This not only distinguishes it from traditional evolutionary theories but also embodies the core essence of the digital product innovation evolution theory.

The proactive selection mechanism is crucial for the rapid innovation and evolution of digital products to adapt to market changes. Meanwhile, the knowledge integration mechanism, which integrates cognitive translation and social translation, is a key difference from traditional product innovation and serves as the critical mechanism for overcoming complex challenges. This mechanism deepens the understanding of the logic behind the innovation process, emphasizing that within the “disorderly” evolution, “orderly” proactive selection and diverse knowledge integration are essential for addressing the uncertainties and complexities in innovation. This holds significant theoretical and practical implications.

Digital product innovation is characterized by complexity. Compared to traditional physical product innovative knowledge, it involves a wider range of problem thresholds and more challenging integration, primarily relying on the complementary fusion mechanism of cognitive and social translation. In the process of digital product innovation, cognitive translation mainly depends on the knowledge and experience of the designers and internal enterprise resources, while social translation emphasizes the extraction of heterogeneous knowledge from external sources. The complementary integration of both facilitates the consolidation and optimization of innovative knowledge.

6.2.3. Drivers of Digital Product Innovation

From the perspective of drivers of digital product innovation, this research uncovered the “secrets” that promote the prosperity of the digital economy. In addition to the traditional pull from customer demand and push from technology, broader cross-industry and fully transparent market competition and the unique driving role of policy guidance with incentives and constraints that have externalities also play a significant role. Additionally, the support from the social digital ecosystem and the digital capabilities of the enterprises themselves also contribute. This expands the scope of the innovation theory’s driving mechanisms and offers a new, broader research perspective.

6.3. Policy and Practical Implications

The theory of digital product innovation evolution emphasizes simultaneous iteration along the dimensions of time and product boundaries to achieve continuous market adaptation and innovation breakthroughs. This theory has significant managerial implications for both enterprises and policymakers, especially given the current acceleration of digital transformation.

For enterprises, the following are involved:

1. **Enhancing Market Adaptability and Strategic Agility:** The proactive selection mechanism requires the management team to have keen market insights and strategic judgment. In the process of digital product innovation, firms need to continuously analyze market demands, competitive dynamics, and technological trends; quickly make strategic adjustments; and choose optimal innovation paths. The cognitive translation mechanism helps companies combine diverse external knowledge with internal experience, transforming it into actual innovation capability. Through these mechanisms, companies can enhance market adaptability and achieve agile strategic adjustments.
2. **Improving Resource Allocation Efficiency and Knowledge Integration Capability:** The proactive selection mechanism allows firms to maximize the use of existing resources and external opportunities within limited resources, improving resource allocation efficiency. The cognitive translation mechanism effectively filters and transforms multi-source information, enabling the integration and application of knowledge from different sources within the company, forming a new knowledge system that continuously fuels innovation. These mechanisms enable firms to efficiently allocate R&D, human, and financial resources, thereby increasing innovation efficiency and success rates.
3. **Building Open Innovation Networks:** Rapidly changing markets require companies to not only rely on internal R&D but also to actively absorb external knowledge and technology. The proactive selection mechanism allows firms to choose suitable external partners, while the cognitive translation mechanism helps them internalize this external knowledge into core competencies. Establishing open innovation networks can enhance the depth and breadth of knowledge sharing and technological cooperation, thereby accelerating the innovation process.
4. **Promoting Organizational Change and Cultural Shaping:** To adapt to constantly changing market demands and technological trends, companies need to undergo corresponding changes in organizational structure and culture. The proactive selection mechanism emphasizes rapid decision making and flexible response, prompting companies to form a flat, flexible, and efficient organizational structure. The cognitive translation mechanism encourages cross-departmental and cross-field knowledge exchange and collaboration, fostering an open and inclusive culture of innovation. Through organizational change and cultural shaping, firms can enhance their innovative capacity and market competitiveness.
5. **Accelerating Technological Iteration and Product Optimization:** In the digital product innovation process, the proactive selection mechanism enables firms to quickly select technological routes and market positions, thereby accelerating technological iteration. The cognitive translation mechanism helps companies merge internal technological accumulation with new external ideas and methods, continuously optimizing product features and user experience. Through these theoretical mechanisms, companies can rapidly respond to market changes and achieve continuous technological and product optimization and upgrades.

For policymakers, the following applies: Digital product innovation is no longer solely the effort of individual enterprises; it exists within a digital network with significant externalities. The Chinese experience offers insights for many developing countries and advanced economies. Strengthening the guidance and incentives of digital industry policies, increasing investment in the digital infrastructure, lowering the costs and risks of incentivizing innovation, promoting the emergence of digital innovation, enhancing innovation efficiency, and driving the prosperity of the digital economy are all recommended. At the same time, it is important to recognize that digital innovation may also bring new issues such as responsibility alienation, technology misuse, algorithm control, and data security. It is therefore necessary to strengthen adaptive governance over platform structures, channels, algorithms, rules, and responsibilities [77].

6.4. Further Research and Limitations

In this paper, a theoretical model of the digital product innovation process through a single-case longitudinal study was created. The model suggests conducting quantitative analyses of the efficiency and effectiveness of proactive selection mechanisms and cognitive translation mechanisms under different contexts, as well as examining the behavioral and psychological factors influencing the decision makers and team members during the process of proactive selection and cognitive translation. Furthermore, it is recommended to validate the economic model assumptions concerning innovation drivers and externalities. Additionally, in this paper, further exploration into the differential performance of the digital product innovation theory is proposed based on adaptive logic when applied to purely digital products versus hybrid digital and physical products.

Using a single-case longitudinal study in China's highly digitized securities and finance industry, in this study, the following limitations were identified:

1. The range and thresholds of uncertainties faced by enterprises engaged in digital product innovation may vary across different levels of economic development, industry types, and production scales.
2. While it is posited in this paper that incremental and breakthrough innovations are two facets of the same coin, this does not account for the technical difficulty and associated risks of innovation, which are important aspects of innovation uncertainty.
3. The specific context of the Chinese scenario in the case study may limit the generalizability of the findings. China, as a large developing country with rapid economic growth, well-developed digital infrastructure, and proactive digital industry policies, presents a context that differs significantly from other mature, developed economies.

Author Contributions: Conceptualization, Z.W.; Methodology, Z.W.; Software, Y.C.; Validation, Z.W., S.W. and C.Z.; Investigation, Z.W. and Y.C.; Resources, Y.C.; Data curation, Y.C.; Writing—original draft, Y.C.; Writing—review & editing, Z.W., Y.C., S.W. and C.Z.; Visualization, Y.C. and S.W.; Supervision, Z.W. and C.Z.; Project administration, Y.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: Author Chun Zuo was employed by the company Sinosoft Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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